

Small Segmented Network

Lab Build & Configuration Notes

These notes walk through building and testing a small segmented network: a router, an L3 core switch, three VLANs (HR, Sales, Management), three access-layer switches, SSH management access, inter-VLAN routing, and NAT to the internet.

1. Network Topology & Physical Connections

Cable modem → RouterA

Modem port 0 (copper, straight-through) connects to RouterA's Gigabit0/0/0 interface. The coax cable connects into the cloud (ISP).

RouterA → Core1

RouterA Gigabit0/0/1 connects to Core1 (3560 core switch), Gigabit1/0/1.

Core1 → HR-Dept switch

Core1 (3650) port Gigabit1/0/2 connects to the HR-Dept switch, Gigabit0/1.

HR-Dept switch → HR PC 1

HR PC 1's Fast Ethernet 0 connects to the HR-Dept switch, FastEthernet0/1.

Core1 → Sales-Dept switch

Core1 port Gigabit1/0/3 connects to the Sales-Dept switch (2960), Gigabit0/1.

Sales-Dept switch → Sales PC 1

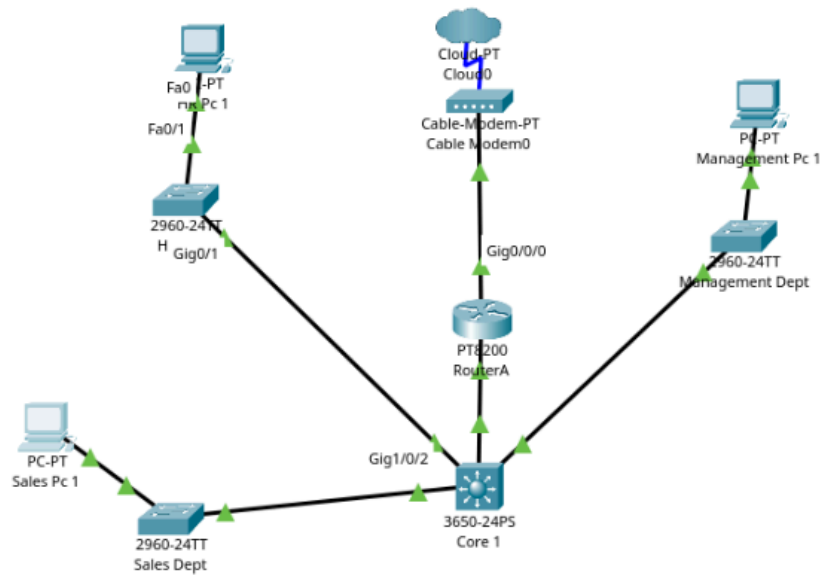
Sales PC 1's Fast Ethernet 0 connects to the Sales-Dept switch, FastEthernet0/1.

Core1 → Management-Dept switch

Core1 port Gigabit1/0/4 connects to the Management-Dept switch (2960), Gigabit0/1.

Management-Dept switch → Management PC 1

Management PC 1's Fast Ethernet 0 connects to the Management-Dept switch, FastEthernet0/1.



Overall topology in Packet Tracer

2. Initial Device Setup

2.1 RouterA

```

enable
configure terminal
hostname RouterA
interface Gigabit0/0/0
ip address 203.0.113.10 255.255.255.0
no shutdown
exit
interface Gigabit0/0/1
ip address 10.1.1.1 255.255.255.252
no shutdown
exit
end
write memory

```

2.2 Core1 (3560 core switch)

```

enable
configure terminal
hostname Core1
interface Gigabit1/0/1
no switchport

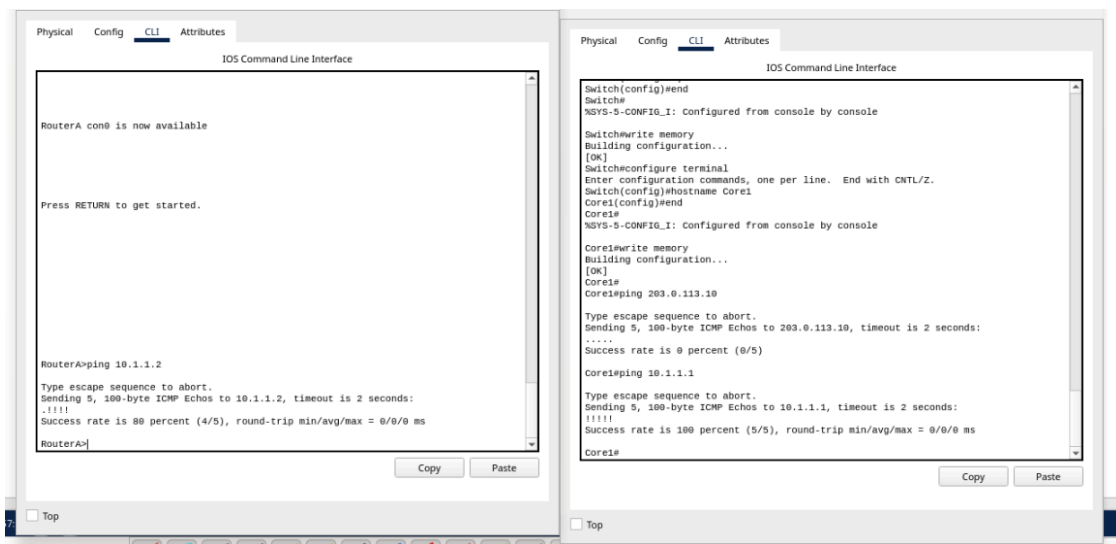
```

```
ip address 10.1.1.2 255.255.255.252
no shutdown
exit
end
write memory
```

TEST

Turn on the 3650 and add the power supply. Ping the IP address connected over the wire to confirm link-level connectivity:

```
RouterA#ping 10.1.1.2
Core1#ping 10.1.1.1
```



Initial ping test between RouterA and Core1

3. VLANs & Trunking on Core1

The 3650 core switch, port 1/0/2, connects to the HR switch, Gigabit0/1.

```
Core1#enable
configure terminal
vlan 10
name HR
vlan 20
name Sales
vlan 40
name management
exit
interface vlan10
ip address 10.10.1.1 255.255.255.0
no shutdown
```

```
exit
interface vlan20
ip address 10.20.1.1 255.255.255.0
no shutdown
exit
interface vlan40
ip address 10.40.1.1 255.255.255.0
no shutdown
exit
ip routing
interface Gigabit1/0/2
switchport mode trunk
switchport trunk allowed vlan 10,40
exit
end
write memory
```

4. HR Department

4.1 HR-Dept switch

```
HR-Dept#enable
configure terminal
hostname HR-Dept
vlan 10
name HR
vlan 40
name Management
exit
interface Gigabit0/1
switchport mode trunk
switchport trunk allowed vlan 10,40
no shutdown
exit
end
write memory
```

TEST

```
show ip route
show vlan brief
show ip interface brief
show interface switchport
```

[Insert screenshot here: HR-Dept show commands output]

4.2 HR PC 1

HR PC 1 connects: Fast Ethernet 0 on the PC to Fast Ethernet 0/1 on the HR-Dept switch.

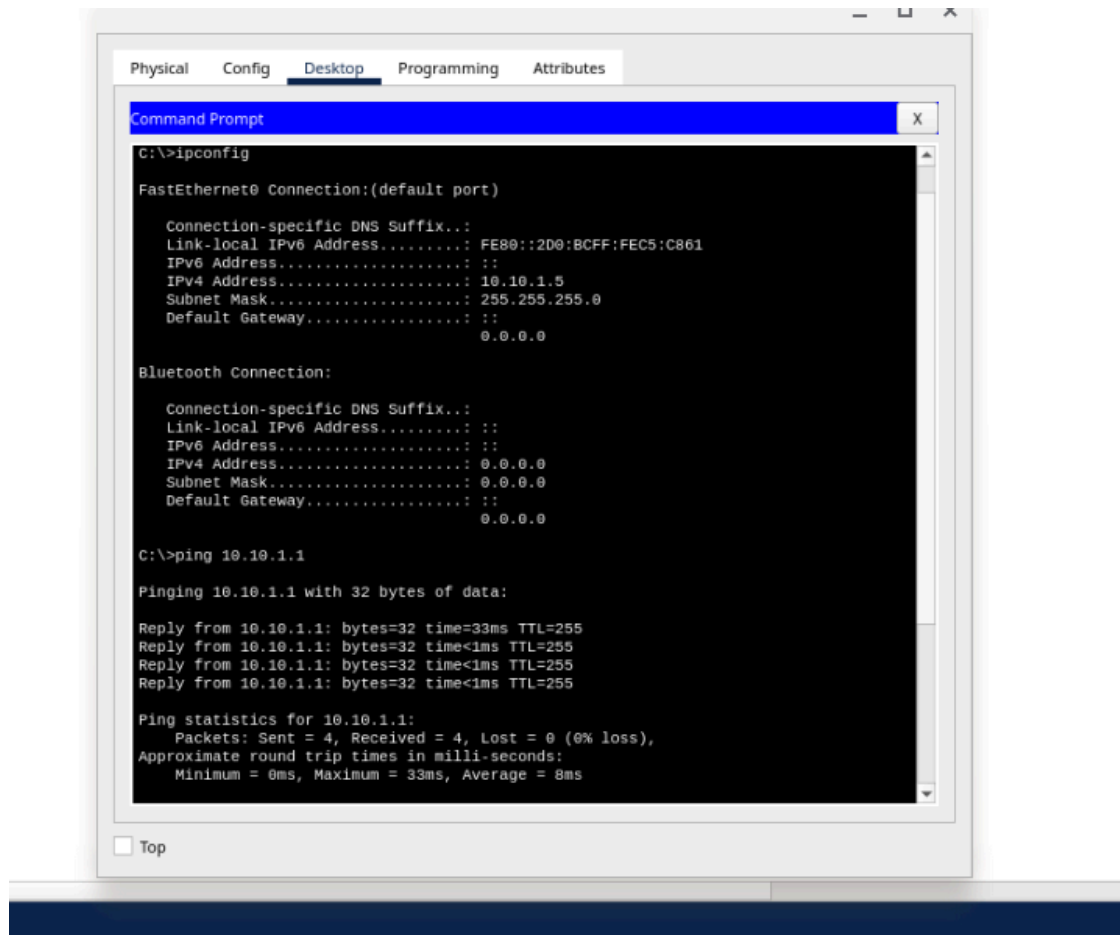
```
HR-Dept# enable
configure terminal
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
no shutdown
exit
end
write memory
```

HR PC 1 static IP: 10.10.1.5 255.255.255.0

TEST

From HR PC 1's command prompt, ping 10.10.1.1 to confirm the PC can communicate with the switch/router. Then ping 10.40.1.1 to confirm it cannot communicate with VLAN 40 from this port.

```
ping 10.10.1.1
ping 10.40.1.1
```



HR PC 1 ipconfig and ping to 10.10.1.1

5. Sales Department

New department, new 2960 switch: Core1 Gigabit1/0/3 connects to the Sales-Dept switch, Gigabit0/1.

```
Core1# enable
configure terminal
interface Gigabit1/0/3
switchport mode trunk
switchport trunk allowed vlan 20,40
exit
end
write memory
```

5.1 Sales-Dept switch

Sales PC 1 connects: Fast Ethernet 0 on the PC to Fast Ethernet 0/1 on the Sales-Dept switch.

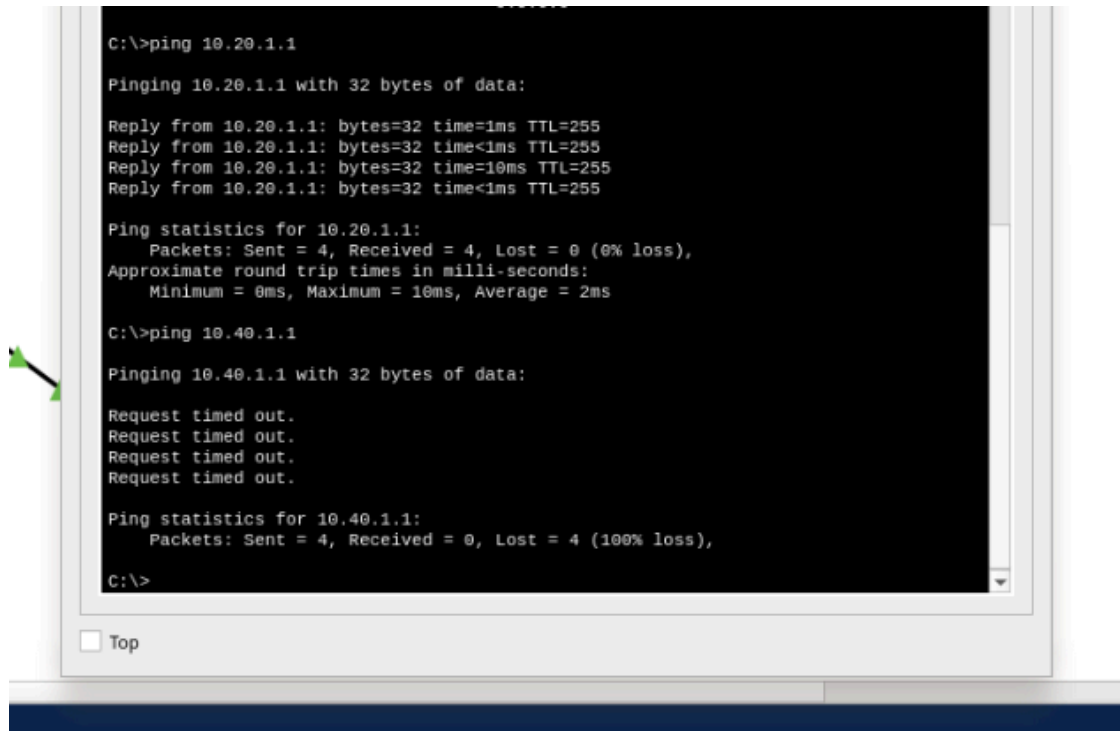
```
Sales Dept# enable
configure terminal
hostname Sales-Dept
vlan 20
name Sales
vlan 40
name Management
exit
interface Gigabit0/1
switchport mode trunk
switchport trunk allowed vlan 20,40
exit
interface FastEth0/1
switchport mode access
switchport access vlan 20
no shutdown
exit
end
write memory
```

Sales PC 1 static IP: 10.20.1.5 255.255.255.0

TEST

Ping 10.20.1.1 (should succeed), then ping 10.40.1.1 to confirm this VLAN cannot communicate with VLAN 40.

```
ping 10.20.1.1
ping 10.40.1.1
```



Sales PC 1 ping test — 10.20.1.1 succeeds, 10.40.1.1 blocked

6. Management Department

New department, new 2960 switch: Core1 Gigabit1/0/4 connects to the Management-Dept switch, Gigabit0/1.

```
Core1# enable
configure terminal
interface Gigabit1/0/4
switchport mode trunk
switchport trunk allowed vlan 40
exit
end
write memory
```

6.1 Management-Dept switch

Management PC 1 connects: Fast Ethernet 0 on the PC to Fast Ethernet 0/1 on the Management-Dept switch.

```
Management Dept# enable
configure terminal
hostname Management-Dept
vlan 40
```



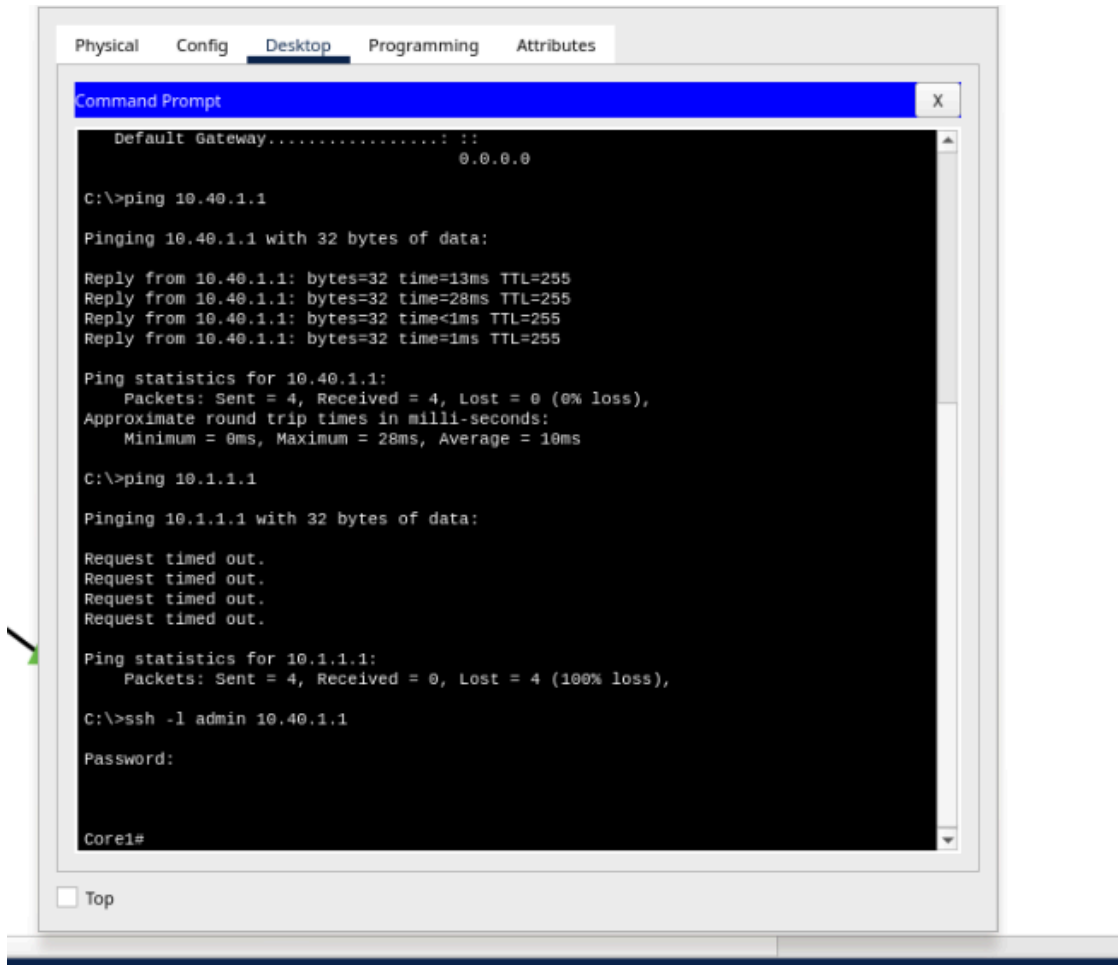
```
name Management
exit
interface Gigabit0/1
switchport mode trunk
switchport trunk allowed vlan 40
exit
interface FastEth0/1
switchport mode access
switchport access vlan 40
no shutdown
exit
end
write memory
```

Management PC 1 static IP: 10.40.1.5 255.255.255.0

TEST

Ping 10.40.1.1 — should succeed.

```
ping 10.40.1.1
```



Management PC 1 ping test to 10.40.1.1

7. SSH Configuration

7.1 Core1

```
enable
configure terminal

! 1. Set the domain name (required for RSA keys)
ip domain-name lab.local

! 2. Create your admin user and password
username admin privilege 15 secret MySecurePassword

! 3. Generate the encryption keys to turn on SSH
crypto key generate rsa
! (When prompted for modulus size, type 1024 or 2048 and press Enter)

! 4. Lock down your VTY lines to use local credentials and only allow SSH
```

```
access-list 50 permit 10.40.1.0 0.0.0.255
line vty 0 4
  transport input ssh
  login local
  access-class 50 in
exit

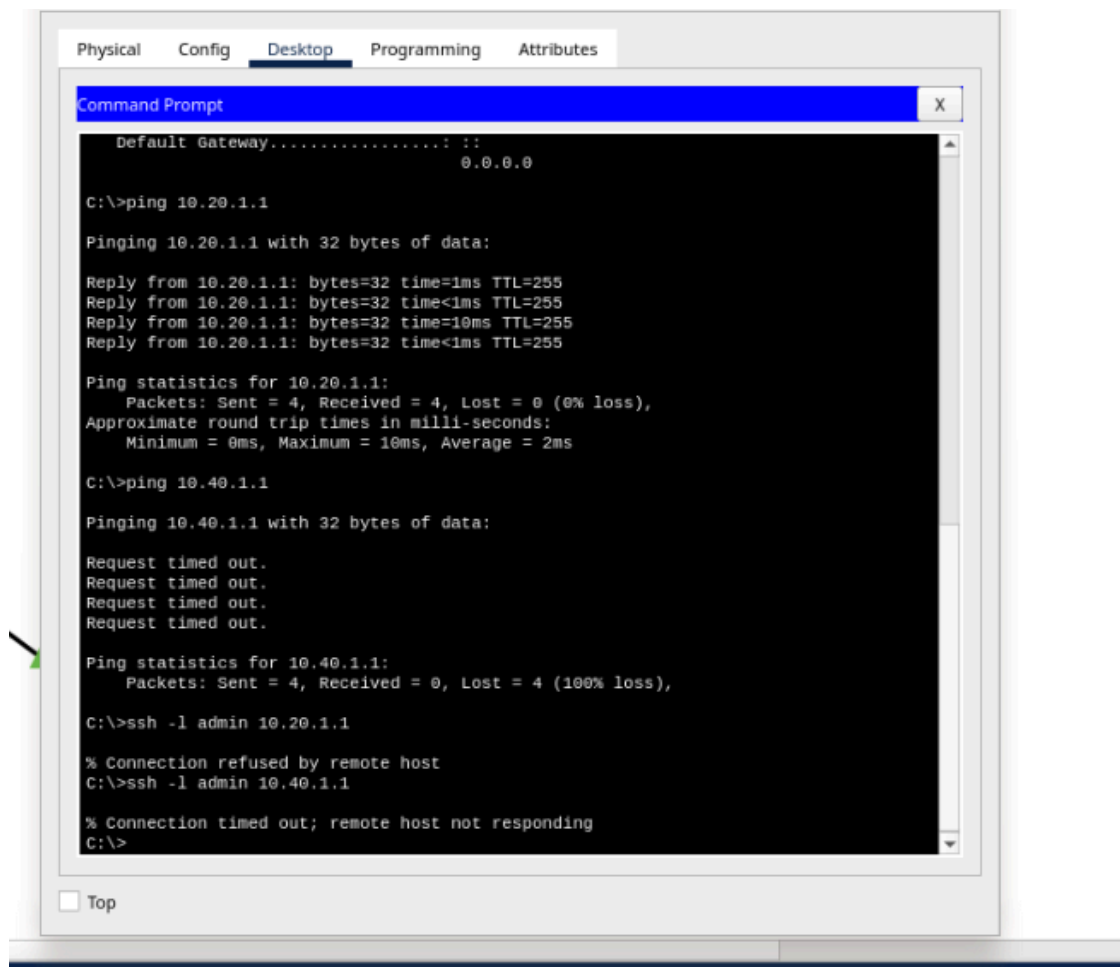
end
write memory
```

TEST

From Management PC 1's command prompt, connect via SSH:

```
ssh -l admin 10.40.1.1
```

Try it from another VLAN as well, to confirm only VLAN 40 is allowed.



SSH to Core1 refused/timed out from the Sales VLAN, confirming only VLAN 40 is allowed

8. Router ↔ Core Communication

```
Router#enable
config terminal
ip route 10.40.1.0 255.255.255.0 10.1.1.2
end
write memory
```

```
Core1#enable
config terminal
ip route 0.0.0.0 0.0.0.0 10.1.1.1
end
write memory
```

Set every PC's default gateway to the Core1 VLAN SVI for that department's VLAN.

TEST

Ping the router from VLAN 40 to test.

8.1 Add the remaining VLAN routes

```
RouterA#enable
configure terminal
ip route 10.10.1.0 255.255.255.0 10.1.1.2
ip route 10.20.1.0 255.255.255.0 10.1.1.2
end
write memory
```

9. SSH for RouterA and the L2 Switches

9.1 RouterA

```
enable
configure terminal

! 1. Set the domain name (required for RSA keys)
ip domain-name lab.local

! 2. Create your admin user and password
username admin privilege 15 secret MySecurePassword

! 3. Generate the encryption keys to turn on SSH
```

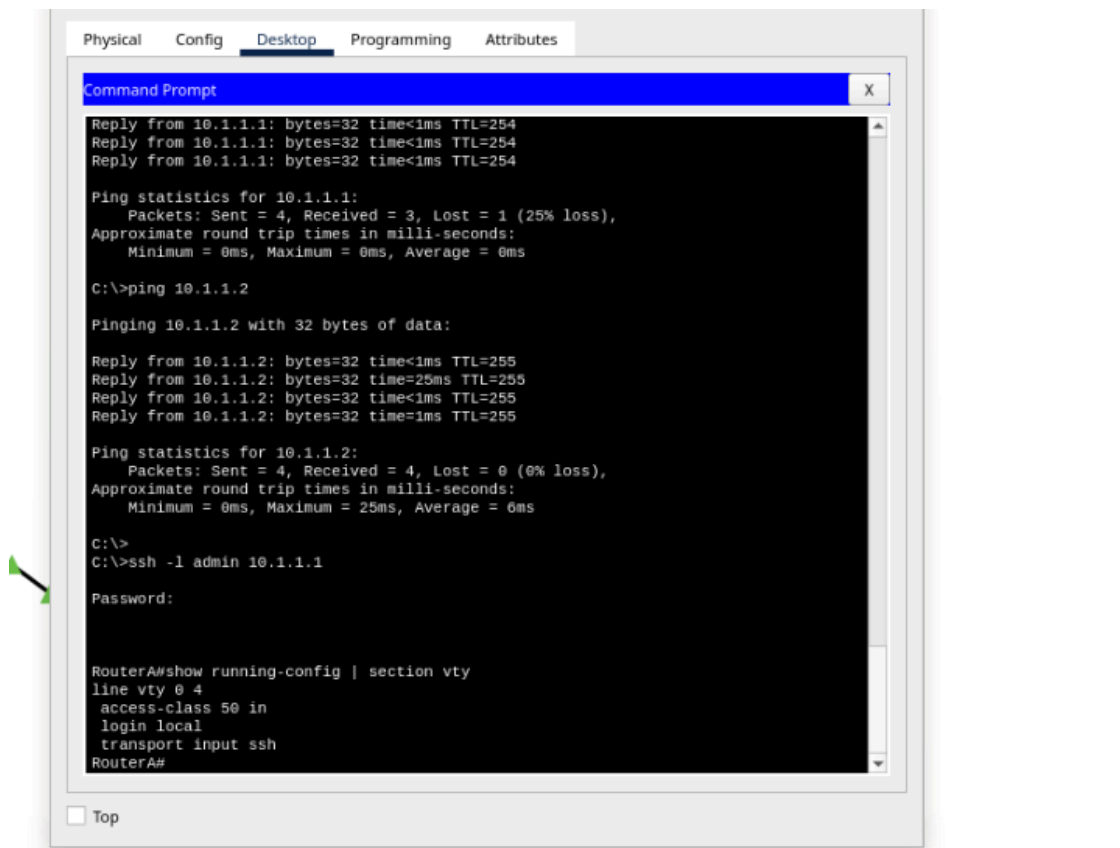
```
crypto key generate rsa
! (When prompted for modulus size, type 1024 or 2048 and press Enter)

! 4. Lock down your VTY lines to use local credentials and only allow SSH
access-list 50 permit 10.40.1.0 0.0.0.255
line vty 0 4
  transport input ssh
  login local
  access-class 50 in
exit

end
write memory
```

TEST

Confirm SSH access and the VTY restrictions on RouterA:



SSH into RouterA (10.1.1.1) and show running-config | section vty

9.2 Repeat for each L2 switch

Give each access-layer switch a management IP on VLAN 40 and point its default gateway at Core1:

```
configure terminal
interface vlan 40
ip address 10.40.1.X 255.255.255.0    (Pick an unused IP in your VLAN 40 subnet)
no shutdown
exit

ip default-gateway 10.40.1.1    (Point this to your L3 switch's VLAN 40 IP)
```

Then enable SSH on each switch:

```
enable
configure terminal

! 1. Set the domain name (required for RSA keys)
ip domain-name lab.local

! 2. Create your admin user and password
username admin privilege 15 secret MySecurePassword

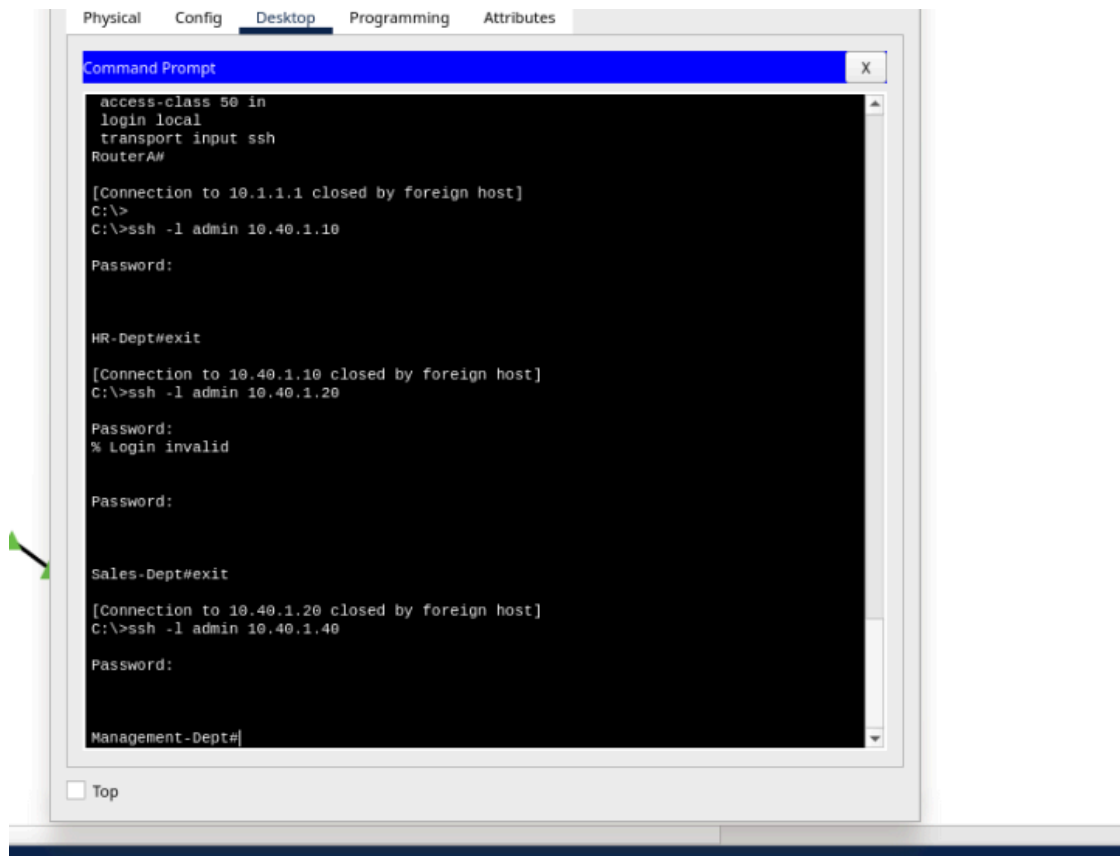
! 3. Generate the encryption keys to turn on SSH
crypto key generate rsa
! (When prompted for modulus size, type 1024 or 2048 and press Enter)

! 4. Lock down your VTY lines to use local credentials and only allow SSH
access-list 50 permit 10.40.1.0 0.0.0.255
line vty 0 4
transport input ssh
login local
access-class 50 in
exit

end
write memory
```

TEST

SSH chained from RouterA into each switch in turn — HR-Dept, then Sales-Dept, then Management-Dept:



SSH session chaining into HR-Dept, Sales-Dept, and Management-Dept

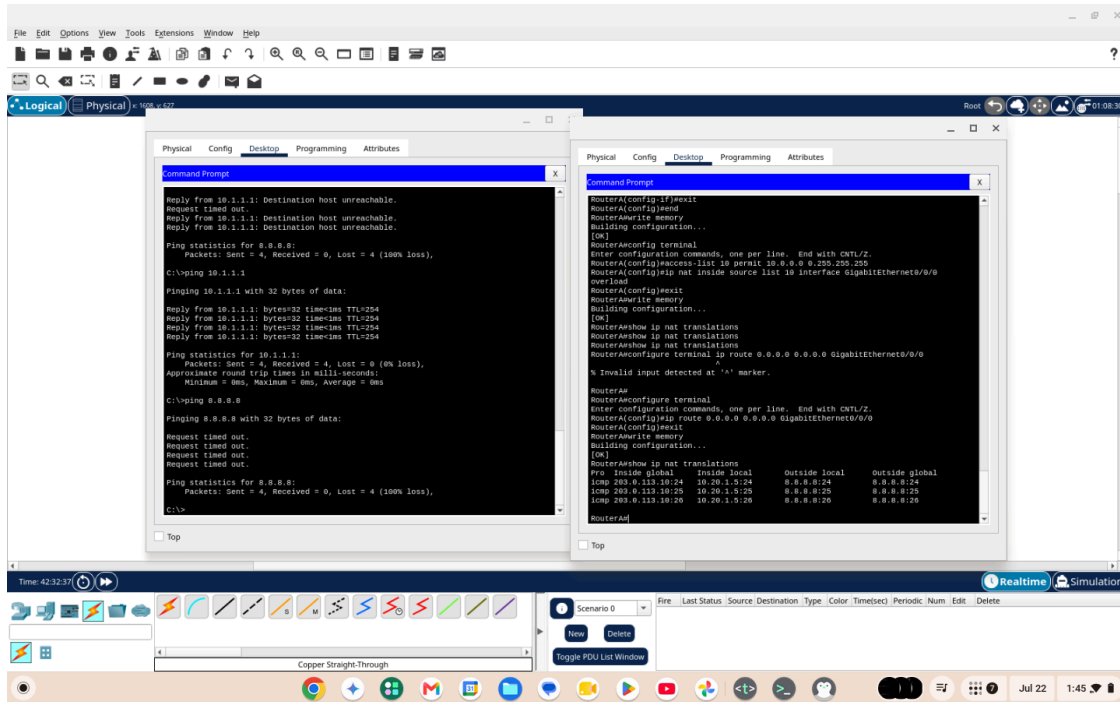
10. NAT Configuration

```
RouterA#enable
config terminal
!Default routing on edge
ip route 0.0.0.0 0.0.0.0 GigabitEthernet0/0/0
interface Gigabit0/0/1
ip nat inside
exit
interface Gigabit0/0/0
ip nat outside
exit
access-list 10 permit 10.0.0.0 0.255.255.255
ip nat inside source list 10 interface GigabitEthernet0/0/0 overload
end
write memory
```

TEST

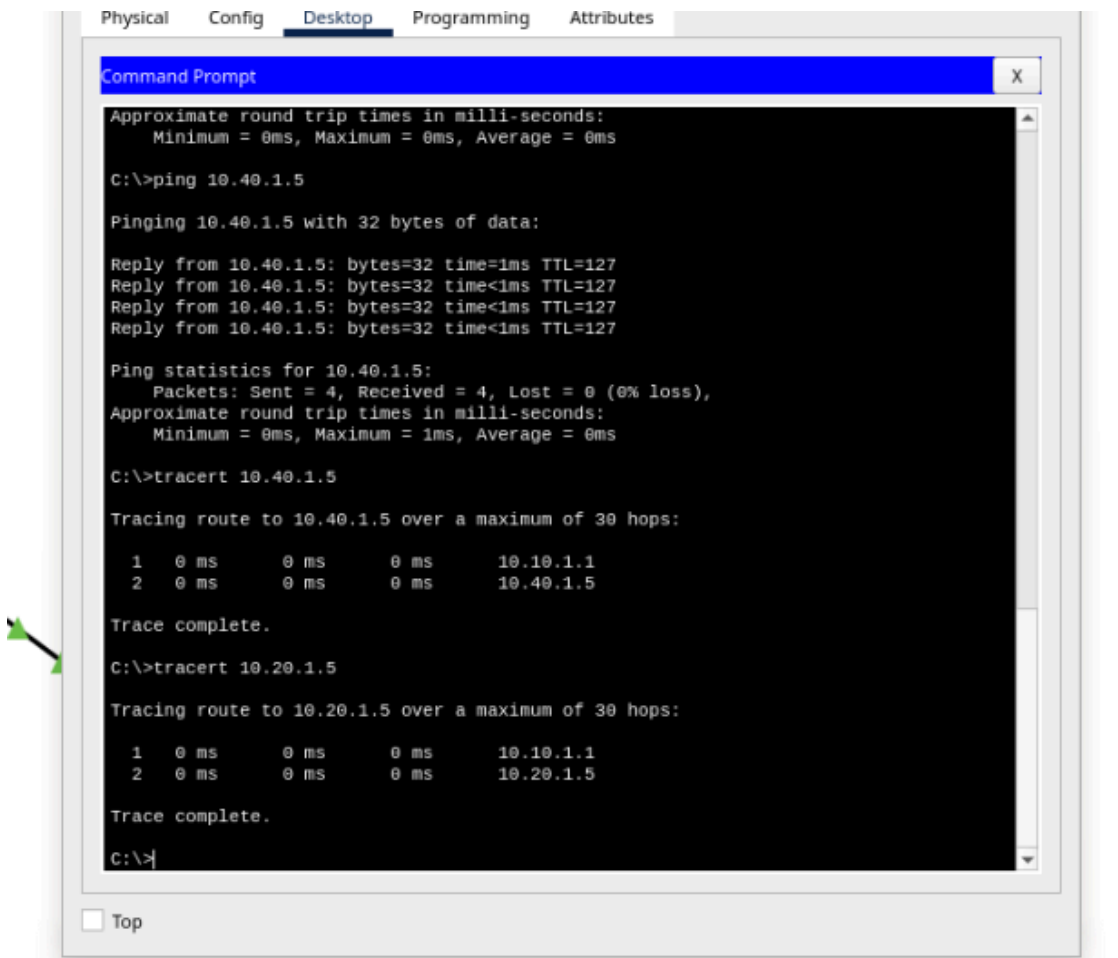
Ping an outside IP address and check the translation table:

show ip nat translations



show ip nat translations — internal hosts translated to 203.0.113.10

Traceroute from different VLANs to confirm traffic is routed through the L3 switch first.



Traceroute from VLAN 40 to Sales and Management PCs, routed through Core1 first